

Artificial intelligence in international business: IB theory under augmented decision-making

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ABSTRACT

This paper discusses the impact of artificial intelligence (AI) on decision-making in multinational enterprises (MNEs) and its theoretical implications for international business (IB) research. Building on the concept of bounded rationality, we analyze how AI affects biases and heuristics that drive human decision-making and identify new, AI-specific biases that could distort decision-making outcomes in MNEs. We propose that AI-augmented decision-making will alter the behavioral assumptions underlying established IB theories, requiring a reevaluation of these theories' boundary conditions. By addressing these shifts, the paper contributes to updating IB research to the realities of AI-augmented decision landscapes.

1. Introduction

New technologies have always been central to international business (IB). Global interconnectedness through the internet (e.g., Kotha, Rindova, & Rothaermel, 2001) has boosted internationalization through seamless communication and e-commerce activities (e.g., Globerman, Roehl & Standifird, 2001), and advanced analytics have shaped the decision-making of international managers (e.g., Duan et al., 2018; Strange & Zucchella, 2017). In this vein, the current revolution in artificial intelligence (AI) and machine learning (Brynjolfsson & McAfee, 2017) will leave its imprint on IB. Especially the augmentation of human decision-making, where decision-makers are supported rather than replaced by algorithm-based technologies, creates new competitive advantages and leads to superior performance (Krakowski, Luger & Raisch, 2022; Raisch & Krakowski, 2021). First applications in industries such as pharmaceuticals (Fleming, 2018) or talent acquisition (Wilson, Daugherty & Bianzino, 2017) provide preliminary evidence for such claims and provide a glimpse into how augmented human decision-making will affect multinational companies (MNEs) and internationalization processes.

However, despite these new opportunities for decision-making and the expected substantial implications of this technological revolution for MNEs, AI applications receive little attention from theoretical IB scholarship. This is a surprising omission because AI can fundamentally

shift decision-making in MNEs. If AI support changes the nature and outcomes of decision-making processes in MNEs, MNEs will change their behaviors. We, consequently, think that IB scholarship needs to incorporate AI into its theorizing. Otherwise, the theoretical perspectives on decision-making in internationalization processes will fall short of the "new" reality that MNEs face.

We argue that the implications of AI applications in MNEs will be strongest for those aspects of theories that relate to behavioral assumptions, specifically those regarding human decision-making under incomplete information. In several prominent theories in IB (and beyond), heuristics and biases stemming from bounded rationality (Simon, 1955) help explain decision-making. Theories like Internationalization Theory, Internalization Theory, Institutional Theory, or the Knowledge-Based View that, at least implicitly, build on the Behavioral Theory of the Firm (Cyert & March, 1963) do not assume that decision-makers operate in a strictly utility-maximizing way. Instead, they suffer from biases that influence their cognitive abilities and rely on heuristics, leading to decisions based on bounded rationality.

We propose that AI applications in MNEs will influence these biases and heuristics in the strategic decision-making processes that are fundamental to IB theories. AI can contribute to lifting the boundaries of rationality that limit MNE managers' decision-making. Consequently, AI helps to reduce how MNE managers suffer from behavioral biases in their decision-making and may challenge assumptions about rationality

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in IB. Put differently, the ongoing AI revolution will shift the established boundary conditions under which IB theories have validity.

Yet, AI and human-AI interaction also suffer from bounded rationality. AI models also have boundaries of rationality, and the resulting machine biases are new and different from well-researched human biases. These machine biases determine how AI shifts the boundaries of rationality. In extreme cases, these machine biases might even result in negative shifts—i.e., MNE managers make more rational decisions without AI than with AI augmentation. As AI-based decision-making becomes a reality in MNEs, IB scholarship must also account for the theoretical implications of these unintended and unexpected consequences.

In this perspective paper, we propose how IB research can acknowledge and incorporate AI into the theoretical mechanisms underlying how we predict MNE actions. The growing integration of AI into MNE decision-making will challenge existing assumptions of the boundaries of rationality (Bettis & Hsu, 2018; Shick, Johnson, & Fan, 2024). Although AI is unlikely to lift all boundaries of rationality, it can help MNE managers become more rational in their decision-making (Ma & Su, 2024). As a result, AI adoption will affect theoretical perspectives where IB researchers have highlighted a major role of biases. We discuss how AI will change biases and heuristics in decision-making routines and what these changes imply for IB theories. In addition, we highlight how machine biases introduced by AI-augmented decision-making require consideration in IB research and affect central IB theories, specifically Internationalization Theory, Internalization Theory, Institutional Theory, and the Knowledge-Based View. We then outline how differences in the institutional environment faced by MNEs condition the implications of AI for decision-making. Based on this understanding of AI-induced change in IB theories, we formulate avenues for future research that we argue are necessary to make IB research ready for an age where most decision-making is, to some extent, influenced by AI support systems. We think that, eventually, the new types of machine biases introduced by AI may even have a stronger effect on IB theory than the impact of AI on traditional behavioral biases.

With this paper, we contribute several important elements to preparing IB research for a future of AI-augmented decision-making. First, building on bounded rationality, we map out how AI will influence decision-making in IB theory. By emphasizing the impact of AI on the boundaries of rationality, we offer a perspective that formalizes the implications of AI for decision-making in IB. Second, we outline how the inherent biases in AI systems could potentially affect decision-making in IB. Our study calls for IB research to account for machine biases in decision-making and to investigate the ability of firms to balance human and machine biases. Third, we discuss how AI-augmented decision-making could affect the mechanisms underlying central IB theories. We suggest that by changing biases and heuristics in decision-making, AI-supported MNEs will make different decisions regarding their international footprint. At the same time, we argue that these changes will unfold contingent on the institutional envelope in which firms operate. We see this as an impetus for IB research to reevaluate the scope and validity of our central theories in a world of human-AI decision-making. Fourth, we outline avenues for future research that will assist IB theory in retaining predictive power over MNE actions in an age of human-AI complementation.

2. Bounded rationality in IB

Neoclassical economic theories assume that individuals make decisions according to rational choices (Nash, 1950; Savage, 1954; von Neumann & Morgenstern, 1944). These theories suggest that individuals will choose among a given, fixed set of alternatives with known probability distributions of outcomes for each alternative to maximize their expected value of a given utility function. For example, when deciding whether to enter a foreign market, an MNE might choose the market that offers the highest expected return based on perfectly rational and

complete information about all potential markets. Thus, decision-makers have complete information and use it to behave rationally. However, in practice, decision-makers cannot know about the full set of available alternatives, the probability distributions of these alternatives, or the exact function underlying their utility generation (Simon, 1990). In addition, humans make their decisions based on their perceptions about their environment (Simon, 2000). For instance, an MNE might perceive a foreign market to be highly risky based on previous experiences in other countries, even though objective data suggests that this new market is stable and offers high potential returns. The bounded rationality framework (Simon, 1956) provides a theoretical representation of how the divergence between the perception of the environment and the actual environment shapes decision-making.

The starting point of the framework is that decision-makers fail to optimize their utility function rationally as they suffer from two boundaries of rationality (Simon, 1955): incomplete information and cognitive limitations. Incomplete information relates to the decision-makers' inability to gather all necessary information for utility optimization. Cognitive limitations relate to decision-makers' inability to process all information for utility optimization. These boundaries of rationality are constraints that increase the differences between the perception of the environment and its objective reality, leading to greater deviation from rational decision-making—i.e., bounded rationality. In the context of FDI, bounded rationality may lead an MNE to misunderstand the difficulties of operating in a new country due to cognitive limitations, preventing it from making optimal investment decisions.

To navigate bounded rationality, decision-makers resort to heuristics—cognitive shortcuts that enable them to make decisions despite incomplete information and cognitive limitations (Gigerenzer, 2004; Simon, 1990). In IB, MNE managers may rely on heuristics when selecting foreign markets for expansion. For example, they favor countries they are more familiar with or have had previous success in, rather than objectively evaluating all potential markets. In this context, heuristics are not mere cognitive quirks but powerful tools that allow decision-makers to operate within bounded rationality.

While heuristics can outperform complex decision-making strategies under conditions of high uncertainty (Artinger, Gigerenzer, & Jacobs, 2022), they also introduce biases, which we understand as systematic errors that subtly influence how information is processed and decisions are made (Dutton & Jackson, 1987; Tversky & Kahneman, 1974). For example, a decision-maker might assume that a market similar to another successful market will perform just as well, leading to overconfidence in market entry decisions. Or, decision-makers might prefer to stick to established markets instead of exploring new opportunities that seem less familiar or riskier.

2.1. Decision-making in IB

A core aspect of IB is that information about foreign markets is generally incomplete. Various IB theories use this lack of information to explain MNE behavior and hypothesize how they may overcome the challenges they face in foreign markets. Due to the prevalence of boundaries of rationality in IB, decision-making under bounded rationality, either explicitly or implicitly, has become a central building block of IB theories (Aharoni, Tihanyi, & Connelly, 2011; Elia, Larsen, Piscitello, 2019; Foss & Weber, 2016; Narula, Asmussen, Chi, & Kundu, 2019; Schubert, Baier, & Rammer, 2018). We focus our discussion of bounded rationality in IB on four theories that are central to IB (while also having applications outside of IB): Internationalization Theory, Internalization Theory, Institutional Theory, and the Knowledge-Based View.

In the context of Internationalization Theory (Johansen & Vahlne, 1977; Vahlne & Johanson, 2017, 2020), bounded rationality affects how MNEs make internationalization decisions (Dow, Liesch, & Welch, 2018; Niittymies, 2020; Vahlne & Schweizer, 2022), select foreign markets (Ambos, Cesinger, Eggers, & Kraus, 2020; Clark, Li, & Shepherd, 2018),

perform R&D internationalization (Belderbos, Leten, & Suzuki, 2013) and capture business opportunities (Sun, Xiao, Zhang, & Zhao, 2018). In particular, risk perceptions and opportunity awareness are critical factors that explain the transformation of experiential learning from earlier commitments to new foreign commitments (Dow et al., 2018). Along these lines, bounded rationality shapes how MNEs use new and potentially contradictory information to decide on internationalization targets and modes. Furthermore, such boundaries of rationality persist in the process of experiential learning in internationalization and can distort learning about foreign markets and business opportunities (Bingham & Eisenhardt, 2011, 2014; Bingham, Eisenhardt, & Furr, 2007; Bingham & Halebian, 2012; Vuori & Vuori, 2014).

Internalization Theory (Buckley & Casson, 1976), along the lines of Transaction Cost Economics (Williamson, 1975), assumes decision-makers suffer from bounded rationality due to incomplete information about the set of a transaction's potential outcomes and likelihoods and cognitive limitations in handling this information (Foss, 2010). Therefore, MNE managers suffer from bounded rationality when choosing the ownership structure in their transactions (Foss & Weber, 2016; Malhotra, Morgan, & Zhu, 2018).

The concept of liability of foreignness and the perspectives on Institutional Theory (which liabilities of foreignness builds on) relate to bounded rationality due to unfamiliarity with the institutional frameworks of foreign countries. The degree to which decision-makers feel familiar with a foreign context plays a crucial role in international investment decisions under Institutional Theory and the liabilities of foreignness perspective. Research suggests that, due to bounded rationality, MNE managers are more likely to invest in domestic markets or "familiar" foreign countries (Bell, Filatotchev, & Rasheed, 2012; De Maeseneire & Claeys, 2012; Filatotchev, Bell, & Rasheed, 2016; Ke, Ng, & Wang, 2010).

Along the lines of the Knowledge-Based View (Kogut & Zander, 1993), bounded rationality can curb the efficient flow (Makela, Andersson, & Seppala, 2012), absorption (Cuervo-Cazurra & Rui, 2017), and sourcing (Monteiro, 2015) of knowledge in organizations. Due to bounded rationality, decision-makers frequently favor existing and proven knowledge within the MNE (Monteiro, 2015).

Our reading of the literature highlights that the central IB theories discuss bounded rationality in a wide array of phenomena across many subdomains. As a result, AI-driven changes in the boundaries of rationality will have a very broad impact on decision-making in IB. These boundaries, in turn, lead to two categories of biases that are particularly prominent in IB research: judgment biases and present state biases. We briefly outline these in the next section before outlining how the use of AI for decision-making will alter the boundaries of rationality and, eventually, change these biases. Afterward, we explain how such changes will affect IB theory. We will also highlight machine biases that AI can introduce into IB theory.

2.2. Heuristics and biases in IB

2.2.1. Judgment biases in IB

Judgment biases¹ arise when individuals must make decisions under conditions of uncertainty—that is, when they lack sufficient information about the objectively correct answer and are thus compelled to rely on subjective impressions (Pohl, 2004). Judgments refer to beliefs about the likelihood of uncertain events, such as predicting the outcome of an election. These beliefs are typically formed using data of limited validity and are processed through heuristic strategies (Tversky & Kahneman, 1974). However, these heuristics or certain characteristics of the event may distort judgment. For instance, individuals may assess probabilities based on how much one event resembles another (representativeness

heuristic), judge likelihood based on how easily examples come to mind (availability heuristic), or anchor their estimates on an initial reference point (anchoring bias). Additionally, people often consider themselves better than average in terms of ability and judgment, reflecting an overconfidence bias (e.g., Alicke & Govorun, 2005; Kruger, 1999).

Judgment biases are critical in shaping key theoretical frameworks such as Internalization Theory and the Knowledge-Based View. These biases become particularly relevant in IB, where managers often face heightened uncertainty due to unfamiliar foreign markets.

2.2.2. Present state biases in IB

Several types of present state biases illustrate how current conditions influence decisions involving information, knowledge, or other parameters. One such bias is the status quo bias, which refers to the tendency to disproportionately favor the existing state of affairs when evaluating alternatives (Samuelson & Zeckhauser, 1988). Another example is the confirmation bias, which involves individuals showing strong confidence in their preferred hypothesis while discounting opposing information (Klayman, 1995; Popper, 1959; Wason, 1960). A further manifestation of present state bias is the familiarity bias, where individuals prefer to make decisions in contexts where they feel more knowledgeable. For instance, decision-makers may favor and allocate resources to more familiar options, even when unfamiliar alternatives might offer greater benefits (Garbuio, King, & Lovallo, 2011; Heath & Tversky, 1991).

Present state biases are highly relevant to understanding internationalization decisions and the behavior of MNEs. For instance, confirmation bias might lead managers to continue investing in markets where they have previously experienced success, even when emerging markets present more profitable opportunities. Similarly, the familiarity effect could cause MNEs to favor countries with institutional frameworks similar to their home country (Bell et al., 2012).

3. AI and bounded rationality in IB

In this section, we outline how the use of AI in decision-making affects the bounded rationality of humans who rely on AI. First, we discuss how AI reduces the boundaries of rationality that humans experience. This is to say, how AI reduces incomplete information and cognitive limitations. Second, we outline how this change to the boundaries of rationality translates into a reduction in the biases that human decision-makers experience when using AI. Third, we propose that AI will not only reduce the boundaries of rationality. AI suffers from machine biases that determine whether and to what degree human-AI decision-making becomes more rational. In this discussion, we use the introduction of AI-driven "control towers" and "digital twins" in Renault's supply chain as an example to illustrate how AI changes decision-making in IB. Fig. 1 summarizes the proposed effects of AI on bounded rationality.

3.1. The effect of AI on the boundaries of rationality in IB

First, AI systems can change how humans make decisions. Human decision-makers often have incomplete information and suffer from cognitive limitations due to insufficient resources regarding processing

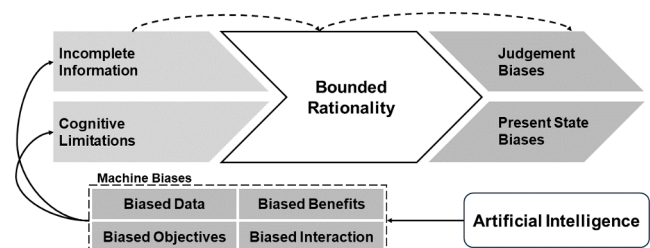


Fig. 1. AI and bounded rationality.

¹ We cluster representativeness heuristics, availability heuristics, anchoring bias, and overconfidence bias under judgment biases.

power and time (Simon, 1956). As a result, decision-makers in MNEs are reliant on heuristics in solving international business problems (Aharoni et al., 2011). Because managers are constrained in the degree to which they can collect and process complex information, MNE strategy designed and executed by humans is naturally based on simplified mental models. AI applications do not have the same limitations as humans regarding processing power and time. Fig. 1 illustrates this effect by the solid arrows leading from AI to the boundaries of rationality.

AI can significantly reduce information gaps in IB by automating the collection and integration of data from diverse and often fragmented sources. Unlike human decision-makers, who are limited in their capacity to continuously scan the environment, AI systems can monitor real-time developments across a wide range of channels, including social media, customer reviews, government publications, and industry reports (Shrestha, Krishna, & von Kroh, 2021). This capability allows AI to detect emerging trends and shifts in market conditions at a speed and scale that would otherwise be unmanageable (von Kroh, Roberson, & Gruber, 2023; Weiser & von Kroh, 2023). For MNEs operating in unfamiliar or dynamic foreign markets, this expanded access to up-to-date, context-specific information enhances market insight and reduces the uncertainty typically associated with international expansion. By drawing on pre-trained models and large-scale datasets, AI tools also enable firms to identify relevant actors, such as potential partners, competitors, or regulators, and assess factors like regulatory complexity, customer demographics, and competitive intensity (Amore et al., 2023). In doing so, AI helps decision-makers build a more complete and nuanced understanding of foreign market environments, thereby addressing one of the key informational constraints underpinning bounded rationality.

Beyond expanding access to information, AI also addresses the cognitive limitations that constrain human decision-making in international business. Managers often struggle to make sense of the vast and complex data required to evaluate foreign markets, especially when that data is unstructured, multilingual, or highly contextual. AI systems help overcome these limitations by processing and structuring such data, for example, extracting insights from local news articles, legal documents, and customer feedback in different languages (Kim et al., 2024). This capability allows MNEs to interpret cultural nuances and institutional conditions that would otherwise be difficult to grasp through manual analysis. In addition, AI tools can identify non-obvious patterns and correlations within large datasets, supporting a deeper understanding of local market dynamics and facilitating the detection of weak signals across global contexts (Grimes, von Kroh, Feuerriegel, Rink, & Gruber, 2023). Explainable AI models further enhance cognitive support by making complex algorithmic outputs more interpretable to human decision-makers. These models help clarify the causal relationships between local developments and global performance, enabling MNE managers to move beyond intuition and engage in more structured, evidence-based reasoning (Senoner, Netland, & Feuerriegel, 2022; Weiser & von Kroh, 2023). In this way, AI not only expands what decision-makers can know but also what they can cognitively process and act upon.

AI can further extend the boundaries of rationality by enabling scenario modeling and real-time feedback, which allow decision-makers to anticipate, test, and refine strategic options with greater precision. Through advanced simulation capabilities, AI can model a wide range of market entry strategies or operational decisions, accounting for variables such as regulatory shifts, supply chain disruptions, and consumer behavior changes across multiple countries (Weiser & von Kroh, 2023). These simulations help MNEs forecast potential outcomes, such as revenue growth, market positioning, or risk exposure, before committing to action. Just as importantly, AI systems provide immediate feedback by analyzing real-time data on how chosen strategies are performing, enabling rapid course corrections. For example, changes in customer sentiment or competitive responses can be quickly detected and incorporated into revised decision models (Lee & Puranam, 2016; Weiser,

Jarzabkowski, & Laamanen, 2020). This continuous feedback loop not only supports more agile and responsive decision-making but also enhances organizational learning by replacing trial-and-error approaches with data-driven experimentation. In complex and dynamic international environments, these capabilities allow firms to adapt more quickly and effectively, thereby reducing the risks associated with uncertainty and misjudgment.

Renault's deployment of AI technologies in its global supply chain illustrates how firms can reduce both informational and cognitive limitations through data-driven decision-making. The company has implemented AI-powered control towers and digital twin systems that monitor real-time data from over 6,000 suppliers, enabling continuous analysis of performance, risk, cost, and sustainability metrics (SupplyChainBrain, 2024; Trecan, 2024). These systems reduce information gaps by scanning vast data sources in real time and highlighting emerging disruptions before they escalate. At the same time, they address cognitive constraints by structuring and synthesizing large volumes of complex supplier data, far beyond what human managers could process alone. The use of digital twins further enhances these capabilities by enabling predictive modeling and scenario testing, helping Renault to visualize future outcomes and make forward-looking decisions (Flaherty, 2022; Renault Group, 2025). As a result, decision-makers in the firm suffer less from incomplete information and cognitive limitations.

3.2. The effect of AI on biases in IB decision-making

Building on our analysis of biases and heuristics in IB research and our understanding of how AI can push the boundaries of rationality, we conceptualize how AI may shift biases that serve as explanatory mechanisms in IB theories. Our central argument is that since AI raises the boundaries humans experience in their rationality, decision-makers will suffer less from bounded rationality and be better positioned to make optimal decisions (Simon, 1955, 1957, 1990, 1997). Consequently, AI will reduce the decision-making biases that result from these biases. Below, we highlight how AI affects the biases that build the basis for many IB theories. Fig. 1 illustrates this indirect effect of AI by the dashed arrows running from boundaries of rationality to bounded rationality and the resulting biases.

AI can play a central role in reducing judgment biases by expanding the range and depth of information available to decision-makers (von Kroh, Roberson, & Gruber, 2023; Weiser & von Kroh, 2023). One key mechanism is AI's ability to generate and evaluate a wide set of alternative scenarios. By simulating multiple outcomes based on diverse inputs, such as regulatory changes, consumer preferences, or competitor strategies, AI enables MNEs to make more informed comparisons between strategic options (Shrestha, Krishna, & von Kroh, 2021). This reduces reliance on the most readily recalled examples or familiar paths, thereby mitigating availability bias. It also weakens the influence of anchoring and overconfidence by challenging initial assumptions and past successes with data-driven projections of unfamiliar contexts (Mussweiler, Strack, & Pfeiffer, 2000; Wang, Yang, Abdul, & Lim, 2019). In addition, AI systems can support more objective reasoning through probabilistic tools. By incorporating base rates (e.g., average success rates of market entrants) and presenting natural frequencies (e.g., likelihood of political instability or currency fluctuations), AI helps decision-makers avoid base rate neglect and reduces representativeness bias, where overly simplistic comparisons between markets may distort judgment (Gigerenzer & Hoffrage, 1995; Kliegr, Bahník, & Fürnkranz, 2020). Real-time simulations and feedback mechanisms further enhance this process by allowing decision-makers to experiment with hypothetical strategies and immediately assess their projected outcomes (Lee & Puranam, 2016; Vert, 2023; Weiser, Jarzabkowski, & Laamanen, 2020). These feedback loops improve experiential learning and reduce over-reliance on intuition or heuristics developed in prior markets. In doing so, AI helps create a more balanced and evidence-based foundation for

strategic decision-making in international business.

AI also helps reduce status quo and present-state biases by encouraging decision-makers to move beyond familiar routines and consider less obvious but potentially more advantageous alternatives (Wang et al., 2019). In IB, managers often favor known strategies or markets due to uncertainty or lack of information about unfamiliar environments. AI challenges this tendency by surfacing data-driven insights that highlight underexplored opportunities, such as emerging markets or unconventional entry strategies, that may not align with past practices but offer strong potential. In doing so, AI reduces the tendency to default to existing preferences simply because they are familiar. Moreover, AI actively counters confirmation bias by presenting disconfirming evidence and incorporating diverse data sources. Instead of reinforcing prior beliefs, AI systems can spotlight contradictory trends, competitor actions, or local dynamics that may challenge an established mental model. This helps decision-makers reevaluate their assumptions and adapt their strategies accordingly. Finally, AI makes unfamiliar or ambiguous environments more navigable by processing and structuring large volumes of unstructured data, such as local news, regulatory texts, or customer sentiment in foreign languages (De Maeseneire & Claeys, 2012). By turning fragmented and context-specific information into coherent insights, AI reduces the cognitive burden associated with interpreting new markets. As a result, decision-makers are less likely to avoid the unknown and more willing to engage with novel, data-informed options.

Taken together, AI's capacity to reduce judgment and present-state biases supports more flexible and adaptive decision-making strategies in international business. By expanding the range of alternatives considered, integrating diverse and disconfirming data sources, and structuring complex or ambiguous information, AI enables decision-makers to move beyond cognitive shortcuts and habitual responses. Importantly, the reduction of bias is not merely cognitive—it is also structural. AI systems embed new decision routines, such as continuous environmental scanning, real-time feedback, and scenario simulation, that institutionalize broader and more balanced information use. These mechanisms help firms move from reactive, experience-based decision-making to proactive, evidence-driven strategy development (Senoner et al., 2022; Wang et al., 2019). As such, AI does not simply enhance decision quality at the individual level but transforms how organizations process information, learn from uncertainty, and engage with global complexity. In this way, AI emerges as a boundary-expanding and bias-countering force that redefines the limits of rationality in IB decision-making.

Renault's use of AI in its global operations offers a practical illustration of how these bias-reducing mechanisms work in tandem. Through AI-powered control towers and digital twin technologies, Renault continuously monitors real-time data from over 6000 suppliers, enabling the simulation of alternative sourcing scenarios and risk forecasts (Flaherty, 2022; Renault Group, 2025; SupplyChainBrain, 2024; Trecan, 2024). These tools reduce judgment biases by allowing decision-makers to test hypothetical outcomes and evaluate supplier performance based on objective, multi-criteria assessments rather than relying on intuitive judgments or past experiences. At the same time, Renault's algorithmic partner selection process helps mitigate present-state and status quo biases by encouraging diversification beyond familiar suppliers. Instead of defaulting to established routines or long-standing relationships, the company uses AI-generated insights to identify strategic opportunities in less conventional geographies and underexplored networks. By embedding these technologies into its decision-making processes, Renault has shifted from reactive, routine-driven practices to a more adaptive and data-informed approach. The company's improved responsiveness to global supply chain disruptions highlights the tangible benefits of using AI to overcome the limitations of bounded rationality in international business contexts.

3.3. Machine biases created by AI in IB decision-making

Above, we outlined how AI helps decision-makers reduce some of their boundaries of rationality and reduce the biases they experience in decision-making. However, AI systems also suffer from their own biases. Like bounded rationality that affects human decision-making, the "rationality" of AI systems can be bounded. This results in machine biases that affect the outcomes of AI-based decision-making by humans. There is a wide-ranging field of research on machine biases that either result from biased training sets or biases in the algorithm itself (e.g., Crawford, 2013; Cowgill & Tucker, 2020; Obermeyer, Powers, & Mullainathan, 2019; Osoba & Welser, 2017). For example, AI has been criticized for reinforcing implicit biases in the training data it obtains, particularly if potentially biased humans classified the data. Crawford and Calo (2016) show that a bias against African-American defendants informed an algorithm designed to predict the likelihood that defendants will re-offend after being released from custody. In another example, Lambrecht and Tucker (2016) show that algorithm-driven targeting of advertisements has an inherent gender bias. By integrating AI support into decision-making, companies import these biases into their decision-making. Table 1 provides a summary of the generic machine biases introduced by applying AI in decision-making. While not intended as a complete list of potential limitations to AI, this section outlines how AI can suffer from some forms of bounded rationality or create new boundaries of human rationality that, in turn, result in new decision-making biases. We group these new AI-driven biases as "biased data," "biased objectives," "biased benefits," and "biased interaction." Biased data and objectives refer to distortions in the functioning of the AI. They are similar to the limits of rationality encountered by human decision-makers and can thus be considered "human-like" machine biases. On the other hand, biased benefits and interaction refer to distortions in the collaboration between humans and AI. These "generic" biases are unique to AI and differ from extant conceptualizations of human bounded rationality. Fig. 1 illustrates these machine biases as a separate box labeled "Machine Biases." Table 1 summarizes the effects of these machine biases.

3.3.1. Biased data of AI in IB

Biased training data in AI may be the category of biases that human decision-makers can relate to most easily. At the heart of the AI algorithm lies a model that often uses enormous amounts of training data. However, limitations in this training data can negatively affect AI systems, which, like human decision-makers, suffer from incomplete information. These limitations may stem from biased human data coding, low data availability, or institutionalized inequalities in the data used (Chandler & Munday, 2011; Karimi, G  nois, Wagner, & Singer, 2018; Ntoutsis et al., 2020; Tufekci, 2014). The incomplete or skewed information in the training data leads to data bias in the AI model (Leavy, Meaney, Wade, & Greene, 2020). For instance, an AI used to assess foreign market attractiveness might disproportionately favor countries with extensive historical data on economic performance (e.g., the United States and Germany) while underestimating the potential of markets with less comprehensive data, such as certain African or Southeast Asian countries. This underrepresentation could discourage MNEs from considering these emerging markets for expansion.

AI systems may also become biased against groups underrepresented in the data or represented only with specific attributes. In the context of international hiring, AI tools trained on hiring data from one region (e.g., North America) might inadvertently introduce biases when used in different cultural or regulatory environments, such as Asia or Europe. Feedback and reinforcement loops in AI training could further solidify this bias (Gebbru, 2020). This bias becomes visible in cases such as targeted advertising, where AI systems might skew their output based on race, gender, or socio-economic status (Lambrecht & Tucker, 2019; Sweeney, 2013). This could cause MNEs to miss significant growth opportunities in markets where the AI's training data does not adequately

Table 1

Overview of machine biases relevant to decision-making in IB.

Machine Biases	Sources	Consequences
Biased Data	Incomplete, biased, or non-representative coverage of “reality” in the training data. This can result from low data availability, human biases in the data collection, or biases affecting the underlying phenomenon.	- AI trained on biased data will propagate the bias in its results. For example, socio-economic bias in the training data results in socio-economic bias in AI-driven advertisements (Neumann et al., 2024). - AI trained on biased data will not generalize to new data and suffer from weak performance. For example, a facial recognition algorithm trained on Caucasian faces will not recognize faces with dark skin tones (Buolamwini and Gebru, 2018).
Biased Objectives	Simplified or incomplete objective functions misrepresent the goals and boundary conditions of an AI’s intended behavior.	- AI will employ “reward hacking” to achieve its objectives with less effort. As a result, the AI will ignore other intended but unspecified goals. For example, an AI programmed to solve all identified problems may refuse to identify new problems (Amodei et al., 2016). - An AI optimized for a simplified objective function will fail to perform satisfactorily when faced with more complex situations. For example, an AI trained to optimize an advertising budget may suffer from unintended gender bias (Lambrecht & Tucker, 2019).
Biased Benefits	The benefits of using AI for the same or similar tasks may vary due to unpredictable capabilities, minor usage differences, and user differences.	- AI may work unexpectedly well on some tasks but underperform on tasks with comparable difficulty. These performance differences are unpredictable in advance (Dell’Acqua et al., 2023). - Minor changes in AI usage (e.g., “prompt engineering”) may result in unexpected performance changes (Meincke et al., 2023; White et al., 2023). - The benefits of AI are not equally distributed across the users’ skill levels. Some users may benefit more from AI support than others (Dell’Acqua et al., 2023).
Biased Interactions	Humans may suffer from biases in their interaction with AI tools. Consequently, the results of AI-based decision-making become biased.	- Due to a lack of trust (Tong et al., 2021) or overconfidence in AI (Bogert et al., 2022), humans may under- or overreact to the results of AI-based recommendations. - Humans may perceive the performance of other humans differently, contingent on whether they use AI (Schweitzer & De Cremer, 2024).

represent local consumer behavior or trends.

From a management perspective, these biases imply that AI-based decision-making could become skewed towards decisions or causal relationships that are overrepresented in the AI’s training data—e.g., a bias towards conservative strategies based on well-established markets, avoiding innovative or riskier strategies in new, fast-growing markets. Similarly, an AI model trained on highly homogenous data from a small subset of markets may struggle to detect black swan events or unexpected opportunities in foreign markets (Weiser & von Kroh, 2023). In such cases, MNEs could miss strategic opportunities due to the AI’s failure to account for non-traditional market dynamics.

3.3.2. Biased objectives of AI in IB

AI, like humans, can suffer from cognitive limitations. While humans are limited by the senses through which they capture their environment, AI is limited by the complexity of its objective function (Amodei et al., 2016). For instance, an AI system to optimize global supply chains might be given a simple objective function to minimize costs. While this may result in cost reductions, the AI could overlook critical factors such as local labor conditions or regulatory requirements, leading to compliance issues or reputational damage for the MNE. Amodei et al. (2016) propose that these distortions can result from the misspecification of the machine’s objective function, negative side effects due to an incomplete objective, or because the AI identifies “shortcuts” to achieve its goals in unintended ways (i.e., reward hacking). In other words, the AI suffers from “incomplete information” about the consequences of its actions or “cognitive limitations” regarding the full complexity of the case to which it is applied.

The resulting biased objectives of the AI can substantially distort AI-based decision-making (Lambrecht & Tucker, 2019). For example, an online algorithm managing career advertisements showed a bias against female website users, targeting them predominantly with product advertisements instead of career-related content. This bias did not arise from data bias (as female users were not underrepresented) but from the AI’s objective function, which prioritized short-term economic value.

3.3.3. Biased benefits from AI in IB

While humans and AI can suffer from bounded rationality that results in biases, these limitations can also extend to the interaction between humans and AI. As a generic AI bias, that differs from human-like biases, biased benefits of using AI manifest in how MNE managers derive value from AI systems in different functional areas. For example, AI might significantly enhance decision-making in supply chain optimization by identifying bottlenecks and reducing costs. However, it may fail to improve decisions in areas where human intuition and experiential knowledge still play a significant role. This uneven benefit of AI usage reflects what Dell’Acqua et al. (2023) describe as a “jagged” technological frontier for AI, where AI improves decision-making for some tasks but not for others. This presents challenges for MNEs in predicting when and how AI will be most useful for specific IB decisions.

Moreover, the growing complexity of AI systems makes it difficult for humans to understand why certain applications work well and others do not. For instance, when MNEs use LLMs for global marketing communications or customer service in different countries, these models evaluate plain-language prompts probabilistically. This makes it challenging for managers to control how the AI responds across different cultural contexts. LLMs might produce unexpected results if a human manager adjusts a prompt by adding emotional or cultural references, which could distort the intended message in a different market (Li, Zhao, Wang, & Liu, 2023). In IB, this unpredictability could lead to inconsistent marketing messages or customer interactions across regions, potentially damaging brand consistency in global markets. This highlights the emerging need for “prompt engineering” (Meincke, Mollick, & Terwiesch, 2023; White et al., 2023) to fine-tune AI interactions, particularly in culturally diverse international settings.

Dell’Acqua et al. (2023) also show that not all human decision-makers benefit equally from AI. While, on average, AI improves decision-making, lower-skilled employees or managers with less international experience may benefit more from AI tools than highly experienced executives who rely on intuition and deep industry knowledge. This unequal distribution of AI benefits can result in biased decision-making outcomes, as those with less experience may become overly dependent on AI, while highly skilled professionals may

underutilize its potential.

Such biased benefits of AI may also be explained by an “overfitting” of human users to AI (Gaessler & Piezunka, 2023). In IB, this might occur when MNEs rely too heavily on AI-based market research, becoming too well-adapted to its patterns and predictions. In this case, the company might struggle when it needs to react to human-driven market shifts, such as sudden geopolitical changes or cultural nuances that AI has not accounted for. As a result, MNE managers might find themselves disadvantaged when faced with unpredictable, human-driven market changes that AI did not prepare them for.

3.3.4. Biased interaction with AI in IB

Research has long discussed the prevalence and implications of technology aversion (Davis, 1989)—a boundary of human rationality in the use of technology. Technology aversion can manifest as resistance to adopting AI tools in global operations. For example, MNE managers may exhibit a status quo bias by resisting AI-driven decision-making tools when expanding into foreign markets, preferring traditional methods based on personal experience and intuition. As another generic machine bias, this biased human-AI interaction can lead to suboptimal outcomes, as managers may undervalue the AI’s capabilities and rely instead on less accurate human judgment. Studies have shown that human decision-makers tend to mistrust AI, even if they experience positive performance effects from AI-based decision-making (Tong, Jia, Luo, & Fang, 2021). Along these lines, incidental observation of AI failure (Dietvorst, Simmons, & Massey, 2015) can undermine confidence in AI systems, leading to biased decision-making.

Trust in AI can differ based on the type of AI, the degree of human involvement in decision-making, or the possibility of adjusting the AI’s outputs. For example, MNEs may trust robotic AI systems to automate repetitive tasks in manufacturing but show hesitation when using AI for higher-level decision-making, such as recommending market entry strategies or pricing models in foreign markets (Glikson & Woolley, 2020). Furthermore, MNE managers might be more willing to trust AI when they have some control or can adjust the system’s parameters, such as fine-tuning AI recommendations based on their knowledge (Dietvorst, Simmons, & Massey, 2018).

However, while distrust in AI can lead to missed opportunities, overconfidence in AI can result in unrealistic expectations. For instance, an MNE might expect the AI to accurately predict consumer behavior across culturally diverse regions, even though the AI may not be fully attuned to local nuances, resulting in lower performance (Bogert, Lauharatanahirun, & Schechter, 2022). In such cases, overreliance on AI may prevent managers from making critical adjustments based on their own expertise, leading to flawed decision-making in complex global markets.

Human-AI interaction may also suffer from human biases about other humans who use AI. If an MNE uses algorithmic management to evaluate employee performance in its global operations, employees who are managed by AI may be perceived as less creative by their peers or managers, leading to lower collaboration or team cohesion (Schweitzer & De Cremer, 2024). This bias can affect the overall perception of AI-managed employees, potentially limiting their career progression or involvement in important decision-making processes within the organization. Consequently, humans’ bounded rationality not only affects their interaction with AI but also indirectly influences how others interact with AI.

3.4. Implications resulting from institutional variation

In addition to the technological constraints that AI-augmented decision-making faces, MNEs will also have to navigate a complex institutional environment when it comes to how AI is used, trained, and governed. We see variation in both formal and informal institutional environments as potential contingencies on how AI shifts the boundaries of rationality.

Regarding formal institutions, variation in the regulatory

frameworks governing AI are at the forefront of potential sources for biases. For example, the European Union’s General Data Protection Regulation (GDPR) imposes rigorous data protection standards, affecting how MNEs collect, process, and analyze data (Brouthers, Geisser, & Rothlauf, 2016; Coche, Kolk, & Ocelik, 2024; Cui, Meyer, & Hu, 2014). When entering markets with robust data regulations, MNEs might experience constraints in using AI for analytics, leading to reduced market responsiveness and competitive disadvantage compared to markets with less stringent data privacy regulations. Conversely, operating in environments with lax regulatory frameworks allows MNEs to leverage more extensive consumer datasets, potentially enhancing predictive capabilities and strategic agility, albeit raising ethical concerns (Gaur et al., 2014). As a result, AI-informed decision-making may favor some markets over others, not because these markets are inherently more attractive, but because AI has an easier time analyzing their data.

Additionally, government investment in AI infrastructure creates disparities in MNEs’ strategic choices across markets. Countries such as China and the United States, which heavily invest in AI research and development, foster environments conducive to rapid adoption and innovation in AI-driven strategic decisions (Buckley, Doh, & Benischke, 2017; Luo & Tung, 2018). MNEs operating in such supportive countries can utilize advanced AI tools for decisions related to global market expansion, risk assessment, and resource allocation. In contrast, MNEs in countries with minimal AI infrastructure support might rely more heavily on traditional decision-making methods, resulting in slower responsiveness to international market dynamics and potential missed opportunities in competitive global markets. As a result, uneven access to AI support may create an uneven playing field between MNEs, or even within MNEs, regardless of product or service quality.

Regarding informal institutions, cultural differences profoundly affect the adoption and integration of AI in international business operations. Societies exhibiting high technology acceptance, such as South Korea and Japan, are more likely to embrace AI-driven decision-making tools across various business functions, including market analysis, product development, and customer relationship management (Magnusson, Westjohn, Semenov, Randrianasolo, & Zdravkovic, 2013; Reiche, Lee, & Quintanilla, 2012). This facilitates smoother implementation and acceptance of AI-enhanced strategic decision-making in MNEs. Conversely, countries characterized by high uncertainty avoidance, prevalent in parts of Europe and Latin America, may show resistance towards AI-based decisions, particularly in sensitive areas like hiring, performance evaluation, or strategic market entry, leading to delayed or suboptimal AI integration. While it is unclear whether AI-supported decision-making is superior to human-only decision-making, uneven use of AI will lead to differences in how decisions are made, and hence investment strategies, for example, might look different for reasons unrelated to economic optima.

Finally, language barriers might further compound cultural variation, as most AI tools are optimized for widely spoken languages such as English, Spanish, or Chinese. AI-driven market analyses and communication strategies can thus perform sub-optimally in linguistically diverse markets, like Eastern Europe, smaller Asian economies, or multi-language countries such as India or Nigeria, where local language nuances significantly impact consumer behavior and market interactions (Brannen, Piekkari, & Tietze, 2014; Harzing & Feely, 2008). For example, AI-driven customer service or marketing tools might inadequately interpret culturally specific idioms, resulting in ineffective marketing campaigns or customer interactions, and consequently weakening an MNE’s competitive positioning and brand perception in these markets. As a result, AI-augmented decision-making may favor more homogeneous markets over more heterogeneous ones.

4. Discussion and propositions

Given the emerging relevance of AI and augmented decision-making

in many functions of international business (e.g., Raisch & Krakowski, 2021; Balasubramanian et al., 2021; Leavitt, Schabram, Hariharan, & Barnes, 2020), it is likely that AI will change decision-making processes and outcomes in MNEs. Research has already provided early evidence that this is the case. For example, Krakowski et al. (2022) argue that AI support changes how companies develop competitive advantage. Similarly, Murray, Rhymer, and Sirmon (2021) argue that when companies include AI in their processes, decision-making changes regarding how humans take agency. Ultimately, it is reasonable to assume that a stronger reliance on augmentation will also change MNE strategizing. In particular, AI, at least partially, lifts and shifts the boundaries of rationality, decreasing behavioral biases. At the same time, AI might itself suffer from human-like and generic machine biases that affect the degree to which MNEs can benefit from AI. While machine biases are likely to lower the benefits of integrating AI in decision-making, in the worst case, they may turn the effect of AI negative. This is to say, the boundaries of rationality increase by using AI. In Table 2 and below, we outline propositions regarding how this could change the assumptions underlying IB theories. We define separate propositions for human biases (Panel A) and for machine biases (Panel B). While we split Panel A into specific theories, the propositions in Panel B apply to IB in general. In Panel C on the bottom of Table 2, we refer to the formal and informal institutions as factors that shape how AI influences IB theories.

The central argument of this study is that AI changes how humans make decisions. Augmentation, i.e., the combination of human and algorithmic abilities, allows human decision-makers to lift and shift their boundaries of rationality. By providing humans with access to new, unstructured, or unknown datasets and massive computing power, AI limits the degree to which humans suffer from incomplete information and cognitive limitations. Consequently, augmented decision-makers suffer less from bounded rationality in their decisions, rely less on heuristics, and their decisions become less distorted by biases. At the

same time, AI creates human-like and generic machine biases that relate to biased data, objectives, benefits, and interactions. In this section, we ask how these AI-induced changes to bounded rationality affect IB theories that describe mechanisms that build on learning and decision-making under uncertainty.

First, we propose how AI-induced changes in decision-making affect IB theories (Table 2, Panel A). We focus on four core theories from IB: Internationalization Theory (P.1-3), Internalization Theory (P.4-5), Institutional Theory (P.6-8), and the Knowledge-Based View (P.9-11), and propose how augmented decision-making will change the mechanisms delineated in the respective theories. The proposed list is not exhaustive, and there might be additional ways by which AI affects the mechanisms suggested by these theories. There are likely also implications for other theories in IB and beyond. We then discuss the prevalence of human-like and generic machine biases, which constitutes an important caveat to the following theoretical implications (Table 2, Panel B).

Internationalization theory. Since AI can improve knowledge sourcing by MNEs, decision-makers will have more available information about the available investment opportunities. At the same time, managers will have more information about the risks related to these investment opportunities. (P.1) Greater knowledge evaluation abilities will allow decision-makers to assess the opportunities and risks for internationalization more accurately and unbiasedly, resulting in better decisions. (P.2) AI-based decision-making will also affect MNEs' learning about their earlier foreign investments. Less biased evaluation of a greater pool of information allows firms to improve experiential learning. Consequently, the link between previous and past commitments will strengthen. (P.3) Scenario planning and experimentation might also offer an avenue to minimize the reliance on costly experiential learning. This will reduce the threshold for firms, particularly those with limited resources and experience, to expand their operations internationally. In

Table 2
Propositions regarding the implication of AI-supported decision-making for theory.

Relevant IB Theories	Propositions Regarding the Implications of AI-Supported Decision-Making	
	<i>Panel A: Human Biases (e.g., Judgment and Present State Biases)</i>	<i>Panel B: Machine Biases</i>
<i>Internationalization Theory</i>	P.1 AI provides decision-makers with more accurate information as well as information that may conflict with previously held beliefs. This helps them to overcome overconfidence and confirmation biases and leads to better internationalization decisions. P.2 AI enhances and creates new learning opportunities, strengthening the link between previous and new commitments. P.3 AI provides inexperienced firms with more accurate information about foreign countries. This reduces availability and familiarity biases and can lower the threshold for these firms' internationalization decisions.	P.12 Biased training data results in algorithms trained on data that are not representative of their fields of application. Consequently, perceptions about foreign countries and the ensuing decisions may all fall short of the complexity of IB. P.13 Biased objective functions may ignore essential boundary conditions. As a result, MNEs will optimize objectives that ignore some of the complexity of IB. P.14 Biased benefits of AI application in MNEs can result in unequal and highly firm-specific effects. Hence, AI's impact on IB theories may not be fully generalizable to all MNEs. P.15 Biased interaction with AI implies that its impact on IB theories is contingent on implementing AI-driven processes in MNEs.
<i>Internalization Theory</i>	P.4 AI reduces biases about the likelihood and materiality of scenarios affecting contracts. This allows the creation of more targeted contracts and reduces the cost of contract creation. In addition, AI facilitates information collection for contract monitoring. Consequently, AI-supported decision-making makes market-based foreign entry more likely. P.5 AI allows firms to enter highly specific contracts (e.g., through product or contractual adaptation). These specific contracts increase the risk of opportunistic behavior, which increases the likelihood of internalization in foreign market entry.	
<i>Institutional Theory and Liability of Foreignness</i>	P.6 AI reduces biases in interpreting information about unfamiliar countries. This facilitates the absorption of new information, which reduces the liability of foreignness experienced by MNEs in these countries. Consequently, domestic and highly experienced firms will benefit less from knowledge advantages. P.7 There will be a gap between AI adopters and non-adopters. While adopters reduce their liability of foreignness, non-adopters will be affected more negatively (since they cannot close their familiarity gap). P.8 New institutions for governing AI will emerge. These institutions will affect whether MNEs can transfer their AI-based processes globally and the degree of adaptation necessary. Non-conformity with these institutions may increase a firm's liability of foreignness.	
<i>Knowledge-Based View</i>	P.9 AI allows firms to access new sources of knowledge and reduces biases in the learning processes that leverage this knowledge. Consequently, AI facilitates the integration of knowledge in global organizations. P.10 AI allows firms to run virtual experiments and conduct real-time causality studies. These new knowledge-generation processes will change knowledge-based competitive advantages. P.11 AI achieves these benefits by making knowledge less context-specific. Consequently, firms can develop new knowledge-based advantages by complementing context-unspecific AI knowledge with context-specific "on-the-ground" learning.	
<i>Panel C: Formal and Informal Institutions</i>	P.16 Formal and informal institutions in home and host countries affect how AI will change human biases and create machine biases. Thus, the implications of AI-supported decision-making are contingent on the institutional envelope in which firms operate.	

conclusion, AI improves learning from past investments, and MNEs can assess opportunities and risks more accurately. Therefore, we assume that AI-supported decision-making reduces the predictive power of core drivers within internationalization theory, specifically psychic distance and time.

Internalization theory. Greater abilities to source and evaluate information brought about by augmented decision-making affect how MNE managers organize firm activities. (P.4) First, managers can identify multiple alternative scenarios to assess the contingencies that may affect a market-based transaction. Second, greater computing power and abilities for knowledge evaluation also allow decision-makers to categorize scenarios. For example, AI lets managers separate critical from uncritical scenarios and likelihood from unlikely scenarios. This, in turn, allows decision-makers to draw up contracts that govern critical scenarios with high likelihoods and ignore uncritical or unlikely scenarios. Third, AI allows managers to design and evaluate complex contracts with provisions for several scenarios and contingencies. Thanks to the lower cost of creating and monitoring these contracts, firms will be more likely to enter market-based transactions rather than internalizing activities. (P.5) However, AI also allows firms to define contracts for highly specialized products and services, which, in turn, raises the threat of opportunistic behavior by partners. Since AI helps to evaluate multiple scenarios, we assume that AI makes contracts more complex and targeted at the most likely critical scenarios. This relaxes (but does not deplete!) one of the core assumptions in transaction cost logic—the assumption of incomplete contracts. At the same time, AI raises the challenge and consequences of opportunistic behavior. In conclusion, we do not assume substantial changes to the outcomes of Internalization Theory. Still, at the same time, augmented decision-making requires a reassessment of the factors driving the internalization of firm activities.

Institutional theory and liability of foreignness. Decision-makers' unfamiliarity with the institutions and conditions in foreign countries is a central topic in applying Institutional Theory and the liability of foreignness. (P.6) AI enhances information sourcing and evaluation, helping managers collect more data about foreign countries and interpret this information without bias. This weakens the familiarity effect favoring the MNE's home country or familiar countries. (P.7) This will lead to a gap between firms that use AI in their decision-making on international expansion and those that do not. While AI-adopters will reduce their liability of foreignness, non-adopters will not be able to reduce their familiarity gap and, consequently, will be affected more negatively by the liability of foreignness. (P.8) However, the increasing role of AI in business and society has also led to the emergence of new institutions that govern the use of AI and data (see the next section below). Non-conformity with these new institutions may increase the liability of foreignness of MNEs. In conclusion, AI increases decision-makers' familiarity with foreign countries and understanding of local institutions, including successful interaction strategies. This weakens the effects of institutions on firms' international strategies.

Knowledge-based view. AI changes how decision-makers source and evaluate information, affecting how organizations structure learning and knowledge dissemination processes. (P.9) AI allows MNEs to use knowledge from multiple data types and structures (e.g., online data, unstructured text data) and different locations (e.g., information collected by subsidiaries). This will facilitate the integration of knowledge in global organizations. (P.10) Moreover, AI enables real-time data collection and causal identification through experimentation. At the same time, AI algorithms enable virtual experiments where "real" participants are replaced by "virtual" ones. These new knowledge-generation processes will change how MNEs create and sustain knowledge-based competitive advantages. (P.11) By making knowledge less context-specific, AI expands the applicability of knowledge and extends learning to new areas. Yet, this also allows firms to generate a knowledge-based advantage by combining context-unspecific AI-based knowledge with context-specific "on-the-ground" learning. In sum, AI-based augmentation will facilitate learning processes and make them

more efficient and less biased. However, these changes will also influence how MNEs create competitive advantages based on sourcing, transforming, and transferring knowledge. In conclusion, based on the profound effects of AI on learning, we assume that AI changes the resources and capabilities necessary to create and sustain a knowledge-based competitive advantage.

Machine biases. The emergence of machine biases is a central implication of AI-based decision-making (Table 2, Panel B). We argue that the implications of these biases are not limited to single theories but are relevant to IB in general. The implications of "human-like" machine biases follow the established lines of human bounded rationality. (P.12) Biased training data can lead to algorithms trained on observations that are not representative of their field of application. This will result in perceptions of foreign countries and decision-making that do not capture the full complexity of IB. (P.13) Biased objectives can lead to simplified objective functions that ignore important boundary conditions and exhibit unintended optimization behavior (Amodei et al., 2016). Hence, MNEs will use AI to optimize decisions in a way that ignores the complexity of IB. However, AI can also create new "generic" biases that are distinct from human boundaries of rationality. Since these generic machine biases deviate from human bounded rationality, their implications for IB theory are less clear. (P.14) Moreover, the benefits of augmented decision-making may differ by application area, application form, and the human decision-maker using AI (Dell'Acqua et al., 2023; Meincke et al., 2023; White et al., 2023). This makes it hard to understand when AI offers the greatest benefit. Therefore, the impact of AI on IB theories may not be fully generalizable to all MNEs and situations. (P.15) Biased human-AI interaction, where humans under- or overreact to reactions from AI, can be a consequence of this (Dietvorst et al., 2015; Tong et al., 2021). In extreme cases, human decision-makers may even become biased against the recommendations of other humans who use AI (Schweitzer & De Cremer, 2024). Thus, the impact of AI on decision-making in MNEs depends on how they implement processes where AI and humans interact.

Formal and informal institutions. It is essential to keep in mind that the implications of AI for decision-making in IB do not unfold in a vacuum. Instead, how AI will shape human biases and whether it will generate machine biases that nullify any positive effects depends on the institutional envelope in which firms operate. Formal institutions like data-governance rules could prevent MNEs from employing AI in some of their host countries (Brouthers et al., 2016; Coche et al., 2024; Cui et al., 2014). Along the same lines, informal institutions like culture or language can also influence the usage of AI for decision-making in IB (Brannen et al., 2014; Harzing & Feely, 2008; Magnusson et al., 2013; Reiche et al., 2012). They might explain why MNEs can successfully leverage AI for decision-making in some countries and will fail in others. (P.16) Consequently, these formal and informal institutions condition the implications of AI-supported decision-making in IB.

In summary, we propose that AI will have fundamental implications for theorizing decision-making in MNEs. However, how AI changes this theorizing also depends on assumptions about MNEs' abilities to handle machine biases. One way of thinking about the tradeoff between human and machine biases in MNEs is to distinguish between automation and augmentation of decision-making (Krakowski et al., 2022). Where AI suffers little from machine bias and can function close to rationality (e.g., in situations with clearly defined rules and unbiased data), managers can use AI to, at least partially, automate decisions to limit the negative impact of human bias. However, AI will augment human decision-making rather than replace it in the face of machine biases. AI helps managers shift their boundaries of rationality, and managers balance machine biases. Consequently, balancing human and machine biases will require hybrid human-AI problem-solving (Raisch & Fomina, 2024). This tradeoff, or rather, firms' resources to balance human and machine biases in decision-making, will become an important boundary condition for IB theories that will require further consideration and investigation.

5. Conclusion and avenues for future research

Revolutionary progress in AI and advances in machine learning have the potential to fundamentally change how managers in MNEs make decisions. Still, despite its efficiency, the integration of AI in decision-making presents several firm-level and societal challenges. At the firm level, AI's inherent biases, originating from training sets and algorithms (e.g., Cowgill & Tucker, 2020; Crawford, 2013), necessitate a mutual de-biasing process with humans to enhance machine learning applications. Intermediary outcomes that aid human understanding of AI suggestions are crucial for effective human-machine collaboration, especially when decisions require human intuition. Additionally, replacing diverse human learning with uniform machine learning might limit organizational knowledge and worsen learning myopia. Societally, ethical dilemmas in AI deployment, like those about security and safety in autonomous vehicles (Gogoll and Müller, 2016), require uniform ethical standards, and the shift towards AI could reshape job skills, particularly in white-collar roles, pushing for rapid acquisition of advanced AI competencies (Kerr and Moloney, 2019). Regulatory measures, such as those under GDPR, emphasize maintaining human oversight in automated systems to ensure accountability and transparency in AI-driven decisions, highlighting the dynamic evolution of legal frameworks in response to AI adoption.

Due to its far-reaching implications and challenges, AI will, like other technological disruptions before, change how MNEs engage in cross-border transactions and make internationalization decisions. Because theories of MNE decision-making are rooted in bounded rationality, AI-induced changes in the boundaries of rationality will influence how MNEs make decisions and how IB theories explain those decisions. Building on the theoretical implications above, we suggest a range of avenues for future research on the micro and macro levels. IB research needs to investigate research questions in these avenues to extend our understanding of MNE decision-making.

First, we need to extend our understanding of how AI changes human biases in the context of MNEs. While behavioral economics has explored a variety of interactions between AI and human biases, few such biases have been investigated in contexts relevant to IB research. Given the complexity of MNEs and the uncertainty within which decisions in IB are made, it remains an open question how much of the behavioral research in clean experimental settings translates into MNEs. For example, although AI can help overcome overconfidence (Kliegr et al., 2020; Senoner et al., 2022), the relationship between overconfidence and internationalization is more complicated. Overconfidence seems to have a non-linear (inverted-U) effect on investment distance (Li et al., 2023), and overconfidence leads to higher stakes in international joint ventures (Lai, Lin, & Chen, 2017). Will AI support similarly change how biases influence investment distance and ownership stakes in joint ventures? Or is it plausible that the overconfidence that substitutes for experiential learning interacts differently with AI than the overconfidence that downplays risk?

Second, we do not know how machine biases will influence MNE decision-making in an IB context, particularly in the case of generic machine biases that deviate from human-like bounded rationality. The complexity inherent in IB contexts may make training powerful AI models more difficult in IB than in other business contexts. While language models for screening CVs can work well in individual contexts, the underlying cultural contexts that play important roles in what is desirable in a candidate differ across countries. AI-supported HR decisions must be more cognizant of machine biases in an international context than in a purely domestic one. Similarly, the institutional context for AI applications differs (Coche et al., 2024; Rong, Ling, Yang, & Huang, 2025), with a particularly challenging context with GDPR in the European Union, which has been at the core of much IB research. Making AI accountable for its decision support (Busuioac, 2021) without over-emphasizing the machine biases will be an important challenge.

Third, major IB theories may make different predictions under

changed boundaries of rationality when humans make decisions with AI support. With more information available via automated data collection and analysis, experiential learning could be partially replaced by other learning types. Still, it is unclear to what extent AI can replace experiential learning in internationalization because a central tenet of experiential learning in internationalization theory is that it needs to happen on the ground and include important contextual information, which AI will have difficulty absorbing effectively. Will we see faster and more distant internationalization in firms that use AI as decision support systems? Similarly, lower uncertainty should lead to cheaper transactions via the market instead of hybrid and hierarchical governance modes. When information becomes more complete, and boundaries of rationality lift and shift to some extent, will we see MNEs opt for more market-based governance modes? Alternatively, it could also be the case that the specificity in AI-supported decision-making leads to a higher expected cost of opportunistic behavior and a positive relationship between AI adoption and hierarchical governance modes. When information becomes easier to absorb, and the institutional context is easier to understand, liabilities of foreignness should recede, and firms should become quicker at catching up with competitors with knowledge advantages. But maybe AI will create a positive feedback cycle, with adopters learning ever more quickly than non-adopters, leading to a change in oligopolistic structures, potentially fewer competitors, and an emergence of duopolies where bandwagon effects or tacit collusion might lead to a reduction in distant market entries. Which one is it going to be?

While we have some insights into AI's linear effects on some biases, the complexities of IB make direct predictions in the light of major IB theories difficult. As a result, we call for research investigating how AI influences MNEs, using all research paradigms and theoretical schools of thought. We have outlined a few particularly salient theories when MNEs adopt AI for decision support. Most new avenues for research will come from interactions between theoretical mechanisms and novel insights into how AI changes and complements human cognition.

CRedit authorship contribution statement

All authors contributed equally to all sections of the manuscript.

Data availability

No data was used for the research described in the article.

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